SHUDDHABROTA BANERJEE

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EDUCATION

Degree / Examination	Institute	Year of Completion	CGPA / %
BTech, Computer Science and Engineering	Delhi Technological University, Delhi	2027	8.3
Senior Secondary (CBSE)	Kendriya Vidyalaya Sec-8, RK Puram, Delhi	2022	96 %
Secondary (CBSE)	Kendriya Vidyalaya Sec-8, RK Puram, Delhi	2020	95.8 %

WORK EXPERIENCE

MLOps Engineer Intern | Scaile.tech (Germany, Remote) *(January '26 – present)*

- Architected and optimized the backend of an **AEO-optimization platform for company context extraction, keyword/blog generation, and AI engine mention analytics**.
- Streamlined **multi-agent workflows, LLM prompts, Redis-based parallel processing,** and caching; integrated Supabase for client data and credits.
- Delivered **2FA authentication and Stripe payments and refined UI/UX**. Deployed the **production-ready full-stack application on Render** for scalable, reliable delivery.

AI Engineer Intern | Matiks *(December '24 – May '25)*

- Developed an **AI-powered multi-agent platform** generating 100+ math question templates with real-time validation; **reduced generation time by 20%** using Vertex AI and created storage with BigQuery.
- Deployed a **Bi-LSTM Deep Knowledge Tracing** model on 20K+ interactions to model topic mastery, **enabling personalized learning paths and adaptive practice.**

PROJECTS

DocumentTranslator | Document Translation & Layout Preservation System | *PaddleOCR,DocLayoutYOLO* | **Github**

- Proposed a dual-pipeline system using **PaddleOCR** for text extraction and **DocLayoutYOLO + PyMuPDF** for layout detection, enabling reliable parsing of multi-page PDFs and scanned documents with strong layout retention.
- Integrated a **Gemini-powered translation agent** with layout-aware prompting to produce coherent, well-aligned translated outputs across 15+ languages, while reducing formatting errors and improving overall translation consistency.

SupRIA-Policy Recommendation system | *Deep Learning, Knowledge Graph, Neo4j* **| Github**

- Built a **Knowledge Graph–powered recommendation system** to streamline insurance policy discovery, reducing user search time by **43%** across a catalog of 1,000+ products through context-aware suggestions.
- Deprecated user-tagging approaches with updated NLP pipelines and reinforcement learning, leading to a 21.3% improvement in prediction accuracy and improved user retention.

CloudGAN | GAN-based Cloud Removal for Satellite Imagery | *PyTorch, CycleGAN, Remote Sensing* | **Github**

- Implemented a **CycleGAN-based model** to translate cloudy Sentinel-2 images into cloud-free counterparts, achieving **0.8 SSIM** and an improved **LPIPS score (0.21)**, reflecting better perceptual similarity across validation scenes.
- Designed the full preprocessing and training pipeline—including cloud-mask filtering, patch extraction, and **AMP-optimized training**—reducing cloud artifacts by **30%** and improving training efficiency by **20%**.

SKILLS

- **Programming Languages:** C , C++, Python, HTML, CSS
- **Libraries & Frameworks:** TensorFlow, PyTorch, OpenCV, Hugging Face, LangChain, LangGraph, LlamaIndex
- **AI/ML:** Vision Transformers, Reinforcement Learning, RAG, BLIP, Knowledge Graphs, XLS-R, Transfer Learning
- **Platforms & Tools:** Google Cloud, AWS, Render, Vercel
- **CS Fundamentals:** DSA, DBMS, CN, SE , Data Warehousing and Data Management
- **DSA:** LeetCode (150+ questions) (**shuban2204**), GFG (**shuddhabrs1et**)
- **Certifications:** Machine Learning, Deep Learning(**Stanford Online**), GenAI with LLMs(**AWS**), Algorithms

ACHIEVEMENTS

- **Grand Finalists in SIH 2025** for the problem statement "Automated Optical Inspection (AOI) based IC marking to identify fake marking".
- **Participant, Amazon ML Summer School 2025** - selected among top candidates nationwide for this program on **advanced Machine Learning and AI**.
- **Grand Finalists** in **SBI LIFE HACK_AI_THON 2024** for the problem statement "Enhancing Customer Experience and Retention through AI-Driven Personalisation to improve the propensity to purchase" from 500+ teams nationwide
- **Winner** at **CodeCrunch ML**, Invictus, Technical Fest, DTU in the **Facial Emotion Detection Challenge by Microsoft** with 70% accuracy.
- **Finalist** at **Adobe Devcraft 2025**, DTU for the problem statement of **Real-time Bidding Optimization.**

POSITIONS OF RESPONSIBILITY

- **Joint Treasurer** of AI and ML society (AIMS-DTU) at DTU (August 2025 – Present)
- **Technical Designer** at ACM-DTU Students' chapter (Dec 2026 – Present)
- **Organizer**, BrAInwave Hackathon, DTU (Oct – Nov, 2024)